

EX. NO. 07 - AFLogical OSE ANDROID DATA EXTRACTION

Experiment Name	Use AFLogical OSE to Extract Data from an Android Device
Aim	To extract logical data such as contacts, SMS, and call logs from an Android device using AFLogical OSE and analyze the collected evidence for forensic purposes.
Tool Used	Android Platform Tools (ADB), AFLogical OSE APK, Google USB Driver for Windows, Android device, USB data cable

OBJECTIVE

- Prepare the required files and experiment folders.
- Connect the Android device to the forensic workstation using ADB.
- Install and run AFLogical OSE to perform logical extraction of supported mobile data.
- Transfer the extracted forensic data to the computer.
- Verify the integrity of selected extracted evidence using SHA-256.

PROCEDURE

Step 1: Prepare the Required Files

Files Required

- Android Platform Tools (ADB)
- AFLogical OSE APK
- Google USB Driver for Windows
- Android device
- USB data cable

Create a main folder for the experiment:

```
C:\ForensicLab
```

Create the following folders:

```
C:\ForensicLab\platform-tools\
```

```
C:\ForensicLab\aflogical-ose\
```

```
C:\ForensicLab\output\
```

Place the downloaded files in their respective folders.

If the AFLogical OSE APK is not available, use the appropriate AFLogical OSE source or a forensic environment such as Santoku Linux.

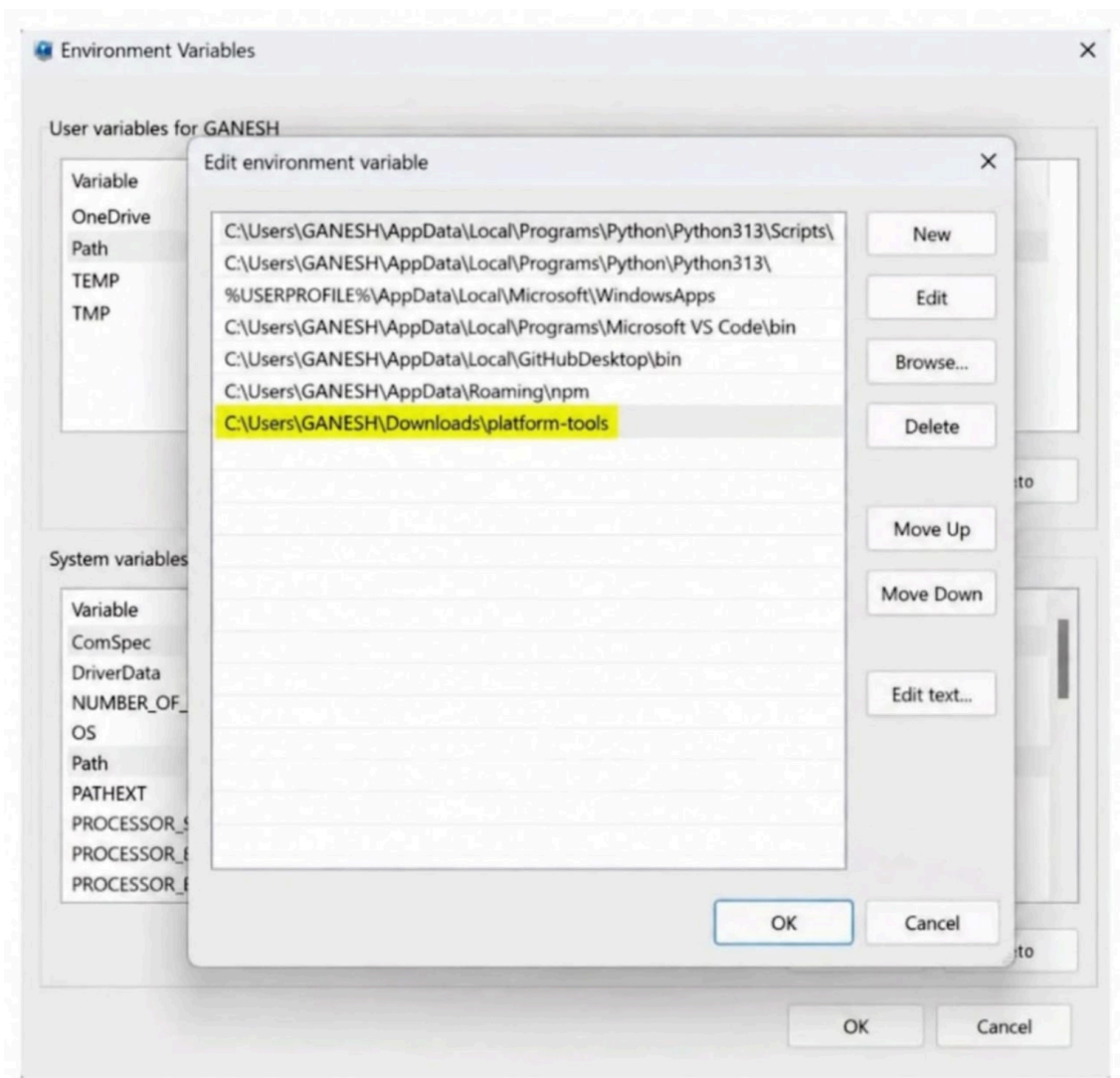
Step 2: Add Platform Tools to PATH

Purpose: To allow ADB commands to be executed directly from Command Prompt.

1. Open: Control Panel → System → Advanced system settings → Environment Variables.
2. Under User Variables, select Path → Edit → New.
3. Add the following platform-tools location: C:\Users\GANESH\Downloads\platform-tools
4. Click OK to save the changes.

Verification

```
adb version
```



Screenshot 1 - Environment Variables

```

Microsoft Windows [Version 10.0.26200.6901]
(c) Microsoft Corporation. All rights reserved.

C:\Users\GANESH\Desktop\df\ForensicLab\platform-tools>adb version
Android Debug Bridge version 1.0.41
Version 36.0.0-13206524
Installed as C:\Users\GANESH\Desktop\df\ForensicLab\platform-tools\adb.exe
Running on Windows 10.0.26200

C:\Users\GANESH\Desktop\df\ForensicLab\platform-tools>adb devices
* daemon not running; starting now at tcp:5037
* daemon started successfully
List of devices attached
10BD541MFE001FN unauthorized

C:\Users\GANESH\Desktop\df\ForensicLab\platform-tools>|

```

Screenshot 2 - ADB version and device verification

Step 3: Install Google USB Driver

Purpose: To enable the Windows computer to communicate with the Android device through ADB.

5. Connect the Android phone to the computer using a USB data cable.
6. Open Device Manager.
7. Locate the connected Android device.

8. Right-click the device and select Update Driver.
9. Select Browse my computer for drivers.
10. Select the Google USB Driver folder.
11. Click Next and complete the installation.

Verification: Open Command Prompt and execute:

```
adb devices
```

Step 4: Enable Developer Options on the Phone

12. Open Settings on the Android phone.
13. Go to About phone.
14. Tap Build number seven times.
15. Return to Settings and open Developer options.
16. Enable USB debugging.
17. Enable Install via USB, if available.

Step 5: Connect the Phone and Check ADB Connection

Purpose: To verify communication between the forensic workstation and the Android device.

18. Connect the phone using a USB data cable.
19. Unlock the phone.
20. Open Command Prompt or PowerShell.
21. Execute:

```
adb devices
```

22. If the phone displays an authorization request, select Allow USB debugging.
23. Run the command again: adb devices

Step 6: Install AFLogical OSE on the Phone

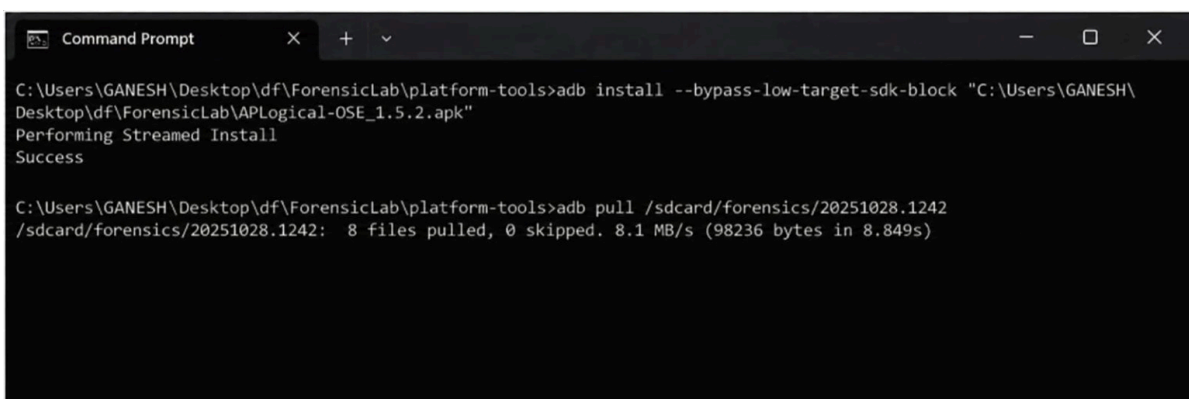
Purpose: To install the forensic extraction application on the Android device.

Ensure that the AFLogical OSE APK is available in: C:\ForensicLab\aflogical-ose\AFLogical-OSE.apk

Open Command Prompt and execute:

```
adb install "C:\ForensicLab\aflogical-ose\AFLogical-OSE.apk"
```

Wait for the installation to complete. The following message indicates successful installation:



```
Command Prompt
C:\Users\GANESH\Desktop\df\ForensicLab\platform-tools>adb install --bypass-low-target-sdk-block "C:\Users\GANESH\Desktop\df\ForensicLab\AFLogical-OSE_1.5.2.apk"
Performing Streamed Install
Success

C:\Users\GANESH\Desktop\df\ForensicLab\platform-tools>adb pull /sdcard/forensics/20251028.1242 /sdcard/forensics/20251028.1242: 8 files pulled, 0 skipped. 8.1 MB/s (98236 bytes in 8.849s)
```

Screenshot 3 - AFLogical OSE installation

Check the Android device and verify that AFLogical OSE is installed.

Step 7: Run AFLogical OSE and Extract Data

Purpose: To perform logical extraction of supported data from the Android device.

24. Open the AFLogical OSE application on the Android device.
25. Grant the required permissions.
26. Select the available data categories for extraction.

- Contacts
- SMS
- Call Logs

- MMS
- MMS Parts

27. Start the extraction process.

28. Wait until the extraction is completed.

Verify the extracted data. The forensic output can be checked using:

```
adb shell ls /sdcard/forensics
```

Step 8: Copy Extracted Data to the Computer

Purpose: To transfer the extracted forensic data from the Android device to the computer for further examination.

Create the output folder:

```
mkdir C:\ForensicLab\output
```

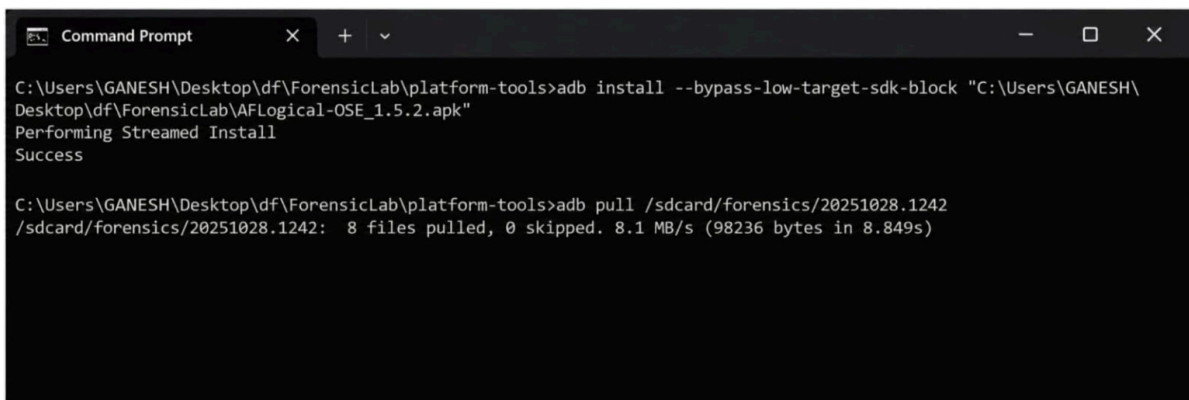
Copy the extracted forensic data using:

```
adb pull /sdcard/forensics/ "C:\ForensicLab\output"
```

Wait for the transfer to complete.

Check the transferred files using:

```
dir "C:\ForensicLab\output"
```



```

C:\Users\GANESH\Desktop\df\ForensicLab\platform-tools>adb install --bypass-low-target-sdk-block "C:\Users\GANESH\Desktop\df\ForensicLab\AFLogical-OSE_1.5.2.apk"
Performing Streamed Install
Success

C:\Users\GANESH\Desktop\df\ForensicLab\platform-tools>adb pull /sdcard/forensics/20251028.1242
/sdcard/forensics/20251028.1242: 8 files pulled, 0 skipped. 8.1 MB/s (98236 bytes in 8.849s)

```

Screenshot 4 - Extracted forensic data transfer

Step 9: Verify Integrity Using SHA-256

Purpose: To verify the integrity of the extracted forensic evidence.

Windows PowerShell Execute:

```
Get-FileHash "C:\ForensicLab\output\<extracted-file>" -Algorithm SHA256
```

Record the generated SHA-256 hash value in the forensic report. Linux/macOS: sha256sum

Step 10: Clean Up

29. Close the AFLogical OSE application.
30. Uninstall AFLogical OSE from the Android device if required.
31. Disconnect the Android device safely.
32. Preserve the extracted forensic files and recorded hash values for documentation.

RUBRICS

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

RESULT

The Android device was successfully connected to the forensic workstation using ADB. AFLogical OSE was installed and used to perform logical extraction of the supported mobile data. The extracted forensic output was transferred to the computer and examined for available contacts, SMS, call logs, MMS, and related information. SHA-256 hashing was used to verify the integrity of the selected extracted evidence.